MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 03 (POLYNOMIALS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM:** 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | $5 \times 1 = 5$ Marks

Q1. If α, β are the zeroes of $p(x) = 2x^2 - 7x + 3$, then $\alpha^2 + \beta^2$ is equal to: [1]

(A) $\frac{37}{4}$ (B) $\frac{49}{4}$ (C) $\frac{25}{4}$ (D) $\frac{13}{4}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

From $p(x) = 2x^2 - 7x + 3$, $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = \frac{3}{2}$. We know:

Formula Used: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Substituting the values:

$$\alpha^2 + \beta^2 = \left(\frac{7}{2}\right)^2 - 2\left(\frac{3}{2}\right) = \frac{49}{4} - 3 = \frac{49 - 12}{4} = \boxed{\frac{37}{4}}. \quad [1M]$$

Q2. A quadratic polynomial with zeroes $5 - 3\sqrt{2}$ and $5 + 3\sqrt{2}$ is: [1]

(A) $x^2 - 10x + 7$ (B) $x^2 + 10x + 7$ (C) $x^2 - 10x - 7$ (D) $x^2 + 10x - 7$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

We know the construction formula for zeroes $\alpha = 5 - 3\sqrt{2}$ and $\beta = 5 + 3\sqrt{2}$:

Formula Used: $p(x) = k[x^2 - (\alpha + \beta)x + \alpha\beta]$

Here $S = (5 - 3\sqrt{2}) + (5 + 3\sqrt{2}) = 10$ and $P = 5^2 - (3\sqrt{2})^2 = 25 - 18 = 7$. Setting $k = 1$ gives $\boxed{x^2 - 10x + 7}$. [1M]

Q3. The number of real zeroes of $f(x) = (x-2)^2 + 4$ is: [1]

(A) 2 (B) 1 (C) 0 (D) 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

We know that the square of any real number is non-negative:

Formula Used: $(x-2)^2 \geq 0 \; \forall \; x \in \mathbb{R} \implies f(x) = (x-2)^2 + 4 \geq 4 > 0$

Since $f(x)$ stays at least 4 and never equals 0 for any $x \in \mathbb{R}$, the number of real zeroes is $\boxed{0}$. [1M]

Q4. If α and β are zeroes of $p(x) = 3x^2 - 6x + 4$, then $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$ is: [1]

(A) 8 (B) 12 (C) 14 (D) 15

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

From $p(x) = 3x^2 - 6x + 4$, we have $\alpha + \beta = \frac{6}{3} = 2$ and $\alpha\beta = \frac{4}{3}$. We know:

Formula Used: $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}, \qquad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$

Evaluating term by term:

$$\frac{2^2 - 2(4/3)}{4/3} + 2\left(\frac{2}{4/3}\right) + 3\left(\frac{4}{3}\right) = \frac{4/3}{4/3} + 2\left(\frac{3}{2}\right) + 4 = 1 + 3 + 4 = \boxed{8}. \qquad [1M]$$

Q5. Assertion-Reason: [1]

Assertion (A): If the graph of $y = ax^2 + bx + c$ does not intersect the x-axis, then the polynomial has no real zeroes.

Reason (R): Real zeroes correspond to the points of intersection with the x-axis.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know the geometric meaning of polynomial zeroes:

Formula Used: x_0 is a real zero of $p(x) \iff (x_0, 0)$ lies on $y = p(x) \cap (x\text{-axis})$

Since the graph does not intersect (or touch) the x-axis at all, there is no real value x_0 with $p(x_0) = 0$. Both **(A)** and **(R)** are true, and **(R)** correctly explains **(A)**. [1M]

Q6. Write the family of quadratic polynomials having $-\frac{1}{4}$ and 1 as its zeroes. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let $\alpha = -\frac{1}{4}$ and $\beta = 1$. We know the general quadratic family formula:

Formula Used: $p(x) = k\left[x^2 - (\alpha+\beta)x + \alpha\beta\right], \quad k \in \mathbb{R} \setminus \{0\}$

Computing sum S and product P:

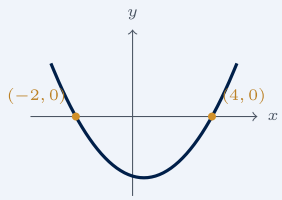
$$S = -\frac{1}{4} + 1 = \frac{3}{4}, \qquad P = \left(-\frac{1}{4}\right)(1) = -\frac{1}{4}. \qquad [1M]$$

Hence the required family is $p(x) = k\left(x^2 - \frac{3}{4}x - \frac{1}{4}\right) = \frac{k}{4}(4x^2 - 3x - 1)$, i.e.:

$$\boxed{p(x) = c(4x^2 - 3x - 1),\ c \in \mathbb{R} \setminus \{0\}} \quad \left(\text{equivalently } k(x^2 - \tfrac{3}{4}x - \tfrac{1}{4}),\ k \neq 0\right). \qquad [1M]$$

Q7. If for a polynomial $p(x)$, the graph of $y = p(x)$ cuts the x-axis at $(-2, 0)$ and $(4, 0)$, write the quadratic polynomial. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Intercept Reading: The x-coordinates of the intersection points $(-2, 0)$ and $(4, 0)$ directly give the two zeroes $\alpha = -2$ and $\beta = 4$. **[1M]**

Since $y = p(x)$ cuts the x-axis at $(-2, 0)$ and $(4, 0)$, the zeroes are $\alpha = -2$ and $\beta = 4$. We know:

Formula Used: $p(x) = k\left[x^2 - (\alpha+\beta)x + \alpha\beta\right], \quad k \neq 0$

With $S = -2 + 4 = 2$ and $P = (-2)(4) = -8$, setting $k = 1$ yields:

$$\boxed{p(x) = x^2 - 2x - 8} \quad (\text{general family: } k(x^2 - 2x - 8),\ k \neq 0). \qquad [1M]$$

Q8. Check whether $g(x) = x - 2$ is a factor of the polynomial $p(x) = x^3 - 3x^2 + 4x - 4$ using the Factor Theorem. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $g(x) = x - 2$ with $x - a$, the zero of $g(x)$ is $x = 2$. We know the Factor Theorem:

Formula Used: $(x - a)$ is a factor of $p(x) \iff p(a) = 0$

Evaluating $p(2)$:

$$p(2) = (2)^3 - 3(2)^2 + 4(2) - 4 = 8 - 12 + 8 - 4 = 16 - 16 = 0. \qquad [1M]$$

Since $p(2) = 0$, by the Factor Theorem, $\boxed{\text{Yes, } g(x) = x - 2 \text{ is a factor of } p(x).}$ [1M]

Q9. If α, β are the zeroes of the polynomial $f(x) = x^2 - 3x - 2$, find the value of: **(i)** $\alpha^3 + \beta^3$ **(ii)** $\alpha^2\beta + \alpha\beta^2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $f(x) = x^2 - 3x - 2$, we have $\alpha + \beta = 3$ and $\alpha\beta = -2$. [0.5M]

We know the cubic symmetric polynomial identity:

Formula Used: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

(i) Value of $\alpha^3 + \beta^3$:

$$\alpha^3 + \beta^3 = (3)^3 - 3(-2)(3) = 27 + 18 = \boxed{45}. \qquad [1.5M]$$

(ii) Value of $\alpha^2\beta + \alpha\beta^2$: Factoring out $\alpha\beta$:

$$\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta) = (-2)(3) = \boxed{-6}. \qquad [1M]$$

Q10. If one zero of the quadratic polynomial $f(x) = 4x^2 - 8kx - 9$ is negative of the other, find the value of k. Hence find the zeroes of $kx^2 + 3kx + 2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the zeroes of $f(x) = 4x^2 - 8kx - 9$ be α and $-\alpha$. We know:

Formula Used: Sum of zeroes $= \alpha + (-\alpha) = 0 \implies -\dfrac{b}{a} = 0$

Here $a = 4$ and $b = -8k$, so:

$$-\frac{(-8k)}{4} = 0 \implies 2k = 0 \implies \boxed{k = 0}. \qquad [1.5M]$$

Finding zeroes of $g(x) = kx^2 + 3kx + 2$: Substituting $k = 0$ directly into the given expression $g(x)$:

$$g(x) = (0)x^2 + 3(0)x + 2 = 2. \qquad [0.5M]$$

Because $g(x) = 2 \neq 0$ for every real number x, it is a nonzero constant polynomial and therefore $\boxed{\text{has NO zeroes.}}$ [1M]

Q11. If α, β are the zeroes of $2x^2 - 4x + 5$, find the value of: **(i)** $\alpha^2 + \beta^2$ **(ii)** $(\alpha - \beta)^2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $2x^2 - 4x + 5$, we get $\alpha + \beta = \frac{4}{2} = 2$ and $\alpha\beta = \frac{5}{2}$ (note $D = -24 < 0$, so symmetric relations apply algebraically). We know:

Formula Used: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \qquad (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

(i) Computing $\alpha^2 + \beta^2$:

$$\alpha^2 + \beta^2 = (2)^2 - 2\left(\frac{5}{2}\right) = 4 - 5 = \boxed{-1}. \qquad [1.5M]$$

(ii) Computing $(\alpha - \beta)^2$:

$$(\alpha - \beta)^2 = (2)^2 - 4\left(\frac{5}{2}\right) = 4 - 10 = \boxed{-6}. \qquad [1.5M]$$

Q12. If the zeroes of the quadratic polynomial $f(x) = x^2 + px + 45$ have a squared difference equal to 144, find the value(s) of p. Also find the zeroes of the polynomial for each value of p. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let α, β be the zeroes of $f(x) = x^2 + px + 45$, so $\alpha + \beta = -p$ and $\alpha\beta = 45$. We know:

Formula Used: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

Step 1 – Finding p: Given $(\alpha - \beta)^2 = 144$:

$$(-p)^2 - 4(45) = 144 \implies p^2 - 180 = 144$$
$$\implies p^2 = 324 \implies \boxed{p = \pm 18} \quad (p = 18 \text{ or } -18). \qquad \textbf{[2.5M]}$$

Step 2 – Zeroes for each value of p:

- **Case 1 ($p = 18$):** $x^2 + 18x + 45 = (x + 15)(x + 3) = 0 \implies$ Zeroes: $\boxed{-15,\ -3}$. **[1.25M]**
- **Case 2 ($p = -18$):** $x^2 - 18x + 45 = (x - 15)(x - 3) = 0 \implies$ Zeroes: $\boxed{15,\ 3}$. **[1.25M]**

Q13. Let α and β be the zeroes of the quadratic polynomial $f(x) = x^2 - 3x - 2$. **(i)** Find the values of $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$. **(ii)** Hence, form a quadratic polynomial whose zeroes are $\alpha^2 + \beta$ and $\beta^2 + \alpha$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

From $f(x) = x^2 - 3x - 2$, we have $\alpha + \beta = 3$ and $\alpha\beta = -2$. We know:

Formula Used: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \qquad \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

(i) Evaluating symmetric powers:

$$\alpha^2 + \beta^2 = (3)^2 - 2(-2) = 9 + 4 = \boxed{13}, \qquad \textbf{[1M]}$$
$$\alpha^3 + \beta^3 = (3)^3 - 3(-2)(3) = 27 + 18 = \boxed{45}. \qquad \textbf{[1M]}$$

(ii) Forming the new quadratic polynomial: For zeroes $\alpha' = \alpha^2 + \beta$ and $\beta' = \beta^2 + \alpha$:

$$S' = (\alpha^2 + \beta) + (\beta^2 + \alpha) = (\alpha^2 + \beta^2) + (\alpha + \beta) = 13 + 3 = 16, \qquad \textbf{[1M]}$$
$$P' = (\alpha^2 + \beta)(\beta^2 + \alpha) = \alpha^2\beta^2 + \alpha^3 + \beta^3 + \alpha\beta = (-2)^2 + 45 + (-2) = 4 + 45 - 2 = 47. \qquad \textbf{[1M]}$$

Using $p(x) = k(x^2 - S'x + P')$ with $k = 1$, the required polynomial is $\boxed{p(x) = x^2 - 16x + 47}$. **[1M]**

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